SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I H.VAISHNAVI(192524251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	5	2.	1	1
Debugging	25%	1	1	1	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

24/10

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26.**

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

(CSA-0707 - 002 - AT3 - Computer Networks) M. VAISHNAV, 192524251

Q1. **IPv4 Debugging**

Given :-

- IP address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default gateway : 192.168.10.1

Ans :-

1. Subnet :- 192.168.10.64/26
2. Network add :- 192.168.10.64
3. Broadcast add :- 192.168.10.127
4. Valid Host Range :- 192.168.10.65
5. IP address valid : = yes
6. Gateway valid No (192.168.10.0/25)
7. configuration Error :- Incorrect default gateway
8. optimized configuration :-
   * IP address :- 192.168.10.65
   * Default gateway : 192.168.10.65
   * Recommended Gateway 192.168.10.126
9. usable Hosts :- 62 ($2^6 - 2$)

Q2 **IPv6 Debugging :-**

Device A :- 2001:db8:ff00:42:8329/64

Expanded : 2001:0db8:0000:0000:0000:ff00:0042:8329

Device B : 2001:db8:0:0:1::1/64

Expanded : 2001:db8:0000:0000:0001:0000
:0000:0001

Network prefix :-

- Device A : 2001:db8:0:0::/64
- Device B : 2001:db8:0:0::/64

Same Subnet ?

Yes, both are in 2001:db8:00::/64

Optimized Addresses:-

Device A : 2001:db8:0D:10/64

Device B : 2001:db8:0:0:20/64

Available Host Addresses:-

- $2^{64} \approx 18.4$ quintillion addresses.

Q3. Subnet optimization :-

Subnet	Network	Host Range	usable host
126	192.168.10	192.168.1.1 -1921681	62
127	192.168.164	192.168.165 - 192.168.1.94	30
128	192.168.196	192.168.197 -192.168.1.10	14

unused address

- /26 -> 62 usable
- /27 -> 30 usable
- /28 -> 14 usable

Inefficient Allocation

- Large department should receive /26
- Medium departments should receive /27
- Small departments should receive /28

Optimized VLSM

- Largest department -> /26
- Small department -> /27
- Medium department -> /28
- Allocate remaining space to future departments

Q4. IPV6 Debugging

Expanded Address A :-

2001:0db8:0000:0000:0000:FF00:8329

Expanded add. B :-

2001:0db8:0000:0000:001:0001

prefix :-

2001:db8:0:0::/64

same subnet : Yes

Available hosts : 2^{64}

Q5. Routing Table Debugging :-

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

No Matching Route Exists :- Default Route

=> 10.0.0.1.